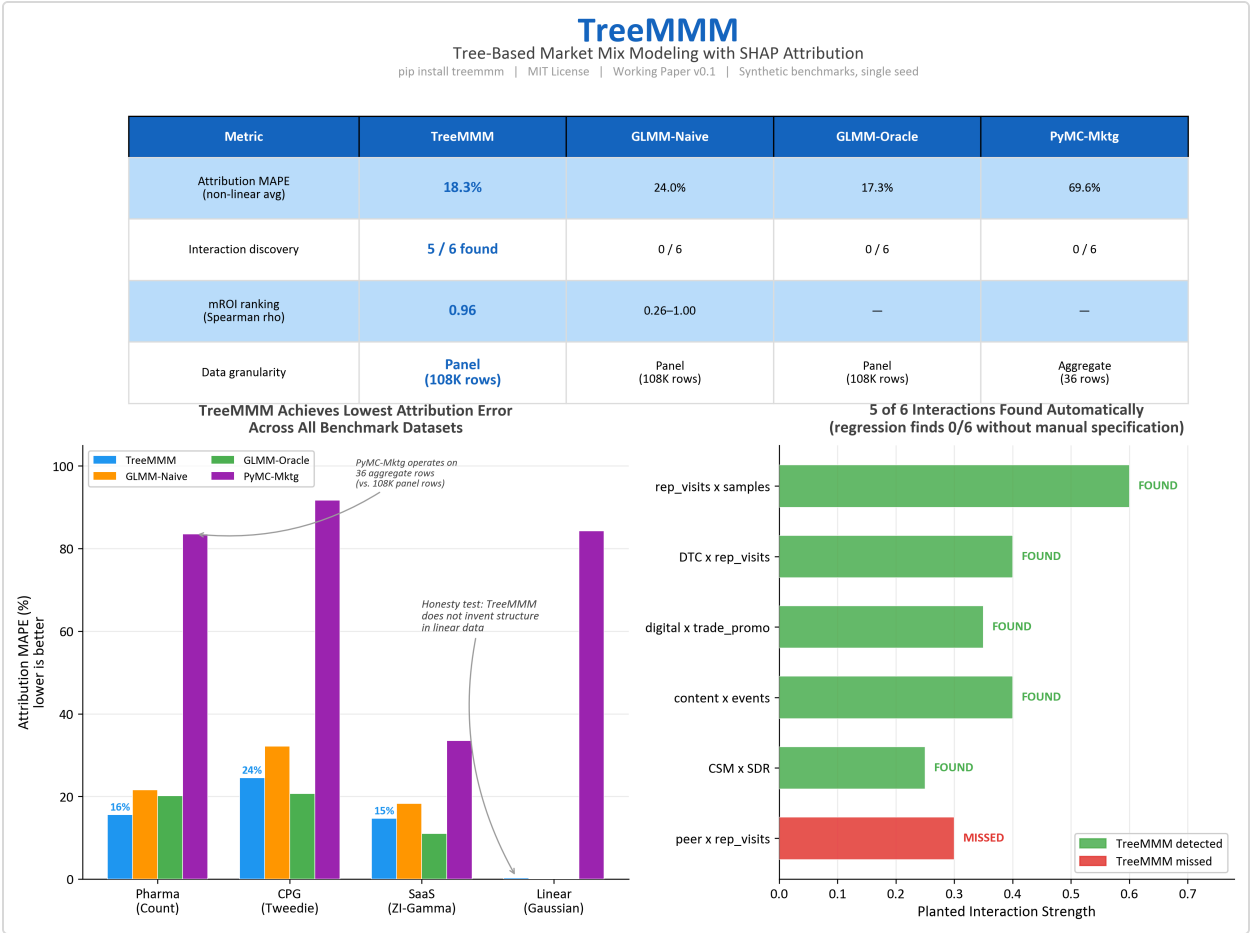


TreeMMM: Tree-Based Market Mix Modeling with SHAP Attribution

James Young, PhD



Visual abstract. TreeMMM compared against GLMM-Naive, GLMM-Oracle, and PyMC-Marketing on attribution share-MAPE across the four benchmark DGPs, plus the interaction discovery summary (5 of 6 planted interactions detected without prior specification).

1. Motivation and Scope

1.1 The “Bayesian MMM is superior” claim is regime-conditional

The conventional framing that Bayesian methods are the right default for marketing-mix modeling was forged in a specific data regime: aggregate weekly time-series with around 150 observations, ten or more collinear channels, and stakeholder demand for posterior intervals. In that regime the conventional framing is correct. Frequentist OLS struggles with weakly identified adstock and saturation parameters, sign-flipping is a real symptom, and priors regularize estimation while allowing informed transfer of prior experimental knowledge. Hierarchical structure benefits from partial pooling. These advantages are real, and they are real because of the data scarcity and aggregation that defines the classical MMM problem.

What changes the calculus is that modern pharmaceutical and B2B promotional analytics rarely produce data in that regime. Promotional data at the healthcare-professional (HCP) panel level (10,000 HCPs and 150 weeks gives roughly 1.5M observations) sits in a different statistical environment. Four panel-data shifts move the right answer away from “default to Bayesian.”

Cross-sectional variation breaks multicollinearity. With 1.5M observations spanning HCPs and time, the rank deficiency that plagues aggregate weekly MMM disappears. The dominant identification path is now within-stratum cross-sectional contrast, not the national-trend separation Bayesian priors are designed to regularize.

Business constraints bound counterfactuals to in-support regions. Min and max touch constraints per HCP force any budget reallocation to redistribute over the existing HCP universe rather than push any individual into unseen territory. Counterfactuals stay within the convex hull of training support, which is exactly where tree-based models are reliable. This is a stronger setup than aggregate MMM saturation curves, which ask the model to extrapolate along a parametric form.

Composite potential variables absorb observable selection bias. Including a well-constructed potential composite (historical prescribing, panel size, specialty, practice type, payer mix, market share trends) means that within strata of potential, variation in promotional intensity approaches as-if-random. XGBoost conditioning on this composite implicitly approximates what double-machine-learning and propensity-score methods do formally. The residual concern, selection on unobservables, is one Bayesian MMM does not solve either.

Out-of-sample validation becomes feasible. With 1.5M panel observations, proper time-series CV produces real OOS prediction guarantees. Aggregate Bayesian MMM with 150 observations has no honest holdout big enough to validate response-curve shape.

Under these four shifts, tree-based methods with SHAP attribution are competitive with or superior to Bayesian MMM for most promotional decisions. Under the classical aggregate regime, they are not.

1.2 What TreeMMM is and is not designed for

TreeMMM is designed for HCP-level (or store-level, account-level, or geo-cell-level) panel data with meaningful cross-sectional variation independent of temporal variation. It is designed for settings where business constraints bound counterfactuals to in-support regions, where observable confounders dominate (rich covariates absorb selection bias), and where the deliverable is response-shape recovery and channel ranking rather than probabilistic statements about a single scalar. It needs sample sizes that support proper time-series CV, in practice at least a few thousand panel-period rows. It is most useful when non-linearities and interactions dominate and the practitioner does not want to specify them by hand.

TreeMMM is not designed for, and the paper does not claim superiority in, several other regimes.

Aggregate weekly time-series with low observation count (around 150 observations across ten or more channels) is Bayesian-MMM territory. TreeMMM has fewer degrees of freedom to spare and no native adstock or saturation parameterization to constrain the search.

Brand-new channels with no historical variation cannot be modeled by a tree. A tree cannot estimate the effect of an input it has never seen at any non-zero level. Bayesian models with informative priors from external lift studies can. TreeMMM does not solve cold-start.

Decisions that require formal probabilistic statements with parameter-level credible intervals are also outside TreeMMM's scope. It provides prediction intervals via conformalized quantile regression and bootstrap, not parametric posterior intervals. If a decision contract requires a 90% credible interval on the rep-visit elasticity, use a Bayesian model.

Selection on unobservables is a problem neither paradigm solves. Where unobserved confounding is the dominant concern, the right tool is experimental design (lift studies, geo experiments, randomized rep visits), calibrating either modeling family. TreeMMM's contribution is in panel-MMM efficiency, not in resolving identification.

Hierarchical sparse-cell estimation is also a poor fit. Where the structure of interest has many small cells (200 HCPs by 50 specialties by 30 payers, with most cells holding fewer than five observations), partial pooling via a Bayesian hierarchical model dominates. Trees fit each stratum independently and overfit on small cells.

1.3 Decision branches

The white paper’s headline claims live on a concrete branch of the decision tree. The branch favors Bayesian structural MMM when most of the following hold: an observation-to-parameter ratio below 20, a design-matrix condition number above 30, treatment variation primarily temporal with channels co-moving (correlation above 0.7), credible informative priors from external evidence, decisions that require probabilistic statements, and hierarchical units with sparse cells. The branch favors a tree-based approach (TreeMMM) when most of the following hold: panel structure with cross-sectional variation independent of temporal, treatment overlap reasonable across covariate strata (propensities away from 0 and 1), counterfactuals that stay within empirical support, confounders that are largely observable, business constraints that bound counterfactuals to in-support regions, nonlinearities and interactions that dominate the response surface, prediction accuracy and OOS validation as deliverables, and a sample size that supports proper CV.

The four pharma DGPs in the paper (pharma_brand, cpg_brand, saas_brand, linear_baseline) are designed to live on the right-hand branch. The linear DGP is included as the honesty test. When the data-generating process is linear and Gaussian, which is the natural home turf of a GLMM or Bayesian regression, TreeMMM should not dominate. The benchmark in Section 3 confirms this. On the linear DGP TreeMMM posts 0.3% attribution-share MAPE against GLMM-Naive’s 0.1%, the expected result for the honesty test.

1.4 Risks of misuse

The paper does not argue that tree-based methods are categorically safer than Bayesian methods. They have different failure modes.

Bayesian-specific failure modes include tight priors centered on desired conclusions, identifiability problems masked by smooth posteriors, MCMC convergence theater (R-hat at 1.01 declared “fine” without further checking), false precision from credible intervals on misspecified models, selective prior reporting, and false confidence in extrapolation along parametric forms.

Tree-specific failure modes include confident-looking SHAP attributions in collinear settings without sensitivity checks, flat extrapolation outside support hidden by smooth predictions,

weak parametric uncertainty quantification, and the temptation to treat prediction accuracy as causal validity.

Shared failure modes include causal claims from observational data without an identification strategy.

Defensible practice in either paradigm comes down to the same set of disciplines, namely showing sensitivity, support, and identifiability for the recovered effects. Section 5 documents which of these checks the paper has executed and which are deferred.

1.5 Diagnostics worth running

Five diagnostics determine whether either modeling paradigm is in defensible territory on a given dataset.

A coverage check counts training observations within a neighborhood of each proposed counterfactual input. If most simulated points have fewer than around thirty nearest neighbors in training, the model is extrapolating regardless of method.

An identifiability check refits with prior variance halved or doubled on the Bayesian side, or with seed and hyperparameter perturbations on the tree side. If channel-level outputs swing more than the decision threshold, the parameters or attributions are not identified.

A treatment-overlap check fits propensity scores for promotional intensity. Common support below 80% across covariate strata means no method recovers causal effects without strong assumptions.

A variation decomposition reports the share of total predictor variance that lives within-unit (temporal) versus between-unit (cross-sectional). The right method depends on where the variation actually lives.

An effective sample size per parameter, computed as `arviz.ess` divided by parameter count on the Bayesian side, or training rows divided by leaves at max depth on the tree side, is informative below roughly twenty observations per parameter, where both paradigms are weakly identified.

Section 5 contains the audit of which of these checks the paper has executed, which are runnable from the package as one-line calls, and which remain follow-up work.

1.6 The hybrid frontier

The right framing for next-generation MMM is not “Bayesian or trees.” It is a question about identification and approximation. What identification strategy fits this data structure, and what function approximator best captures the response surface within that strategy?

Promising hybrids include Double Machine Learning (Chernozhukov et al. 2018) with XGBoost as the nuisance estimator inside an orthogonalized causal framework. This is the most honest path for tree-based methods to reach causal claims rather than predictive ones. Bayesian models with ML-derived priors are another route: use a tree to suggest saturation and adstock parameters, then fit a Bayesian model with informative priors centered on the tree’s estimates and uncertainty bands wide enough to discipline. Geo experiments and randomized rep tests calibrate either modeling class, with identification coming from the experiment rather than the model. The Tree-to-GLMM hybrid introduced in this paper is a simpler variant of the same idea, where the tree mines interactions and the smooth GLMM fits them with spline bases and per-customer random intercepts.

This paper contributes the panel-MMM tree-based building block. It does not claim to resolve the broader identification debate.

2. Methods

2.1 Distribution-Aware Objective Selection

TreeMMM supports four objectives matched to outcome distributions:

Distribution	Objective	When to Use
Gaussian	MSE	Continuous, symmetric (revenue, value sales)
Poisson	Log-link	Non-negative counts (Rx, orders, NPS)
Tweedie	Log-link	Zero-inflated continuous (revenue with stockouts)
Gamma	Log-link	Strictly positive continuous (per-transaction revenue)

An automated diagnostic examines discreteness, zero-inflation rate, skewness, and the mean-variance relationship to recommend an objective. The user can override this recommendation.

2.2 Link-Function-Aware Attribution Decomposition

SHAP TreeExplainer computes values in margin space:

- **Identity link** (Gaussian): $E[y] + \text{sum}(\text{SHAP}_i) = y_{\text{hat}}$. SHAP values are directly additive on the response scale.
- **Log link** (Poisson/Tweedie/Gamma): $\text{base_value} + \text{sum}(\text{SHAP}_i) = \log(y_{\text{hat}})$, where base_value is the mean model prediction in log space. SHAP values are additive on the log scale.

For log-link models, naive exponentiation breaks additivity: $\exp(a + b) \neq \exp(a) + \exp(b)$. TreeMMM's decomposer uses **unsigned proportional allocation**:

$$\text{attribution}_i = (|\text{SHAP}_i| / (|\text{base}| + \text{sum}(|\text{SHAP}_j|))) * y_{\text{hat}}$$

This guarantees: 1. All attributions are non-negative (standard MMM practice: “X% of sales attributed to Lever A”) 2. Attributions sum to the predicted outcome for every observation 3. The relative ranking of channels is preserved from the log-space SHAP values

A unit test verifies this sum-to-prediction property for all four objectives.

2.3 Temporal Cross-Validation

TreeMMM uses time-respecting validation to prevent future data leakage:

- **Rolling origin**: Training window grows forward; test window is the next period
- **Period-jump-forward**: Training window grows; test window jumps by a fixed stride

The minimum training fraction is configurable (default: 50% of periods). No observation from a future period ever appears in the training set.

2.4 Hyperparameter Tuning

Optuna Bayesian optimization tunes tree hyperparameters within each CV fold: - Number of leaves, learning rate, regularization (L1/L2), feature/bagging fraction, minimum child samples - The tuning objective is distribution-matched: Poisson deviance for counts, Tweedie deviance for zero-inflated outcomes, MSE for Gaussian

2.5 mROI Simulation with Extrapolation Safety

TreeMMM’s mROI (marginal Return on Investment) simulator estimates response curves with a critical safety constraint:

- **Per-customer caps** are set at the observed percentile (default: 95th)
- Higher aggregate engagement is achieved by **spreading to more customers**, not pushing any individual beyond observed bounds
- Every customer-level prediction stays within the training distribution

This design ensures the tree model is never asked to extrapolate beyond the feature space it was trained on. The constrained optimizer (scipy SLSQP) surfaces which customers have untapped marginal sensitivity, revealing that “see your top customers the most” (status quo targeting) is often suboptimal.

2.6 Baseline Models

We compare TreeMMM against three baselines representing different modeling paradigms:

GLMM-Naive: Main effects only, random intercepts per customer (statsmodels MixedLM). Represents a typical analyst who does not specify interactions or distributional families.

GLMM-Oracle: Correctly specified interaction terms matching the DGP (statsmodels MixedLM). Represents an analyst with perfect knowledge of which variables interact, but without distributional family matching.

Both GLMM configurations use statsmodels MixedLM, which fits a Gaussian linear mixed model. For the linear dataset, the GLMM is fitted on the raw outcome. For non-Gaussian datasets (pharma, CPG, SaaS), the benchmark log-transforms the outcome variable before fitting, which approximates a log-normal model. This is a common workaround; while statsmodels does offer `PoissonBayesMixedGLM`, it uses a variational Bayes approximation that is not directly comparable to the frequentist MixedLM, and Tweedie/Gamma mixed models are not available. A properly specified distributional GLMM (e.g., `glmmTMB` in R) is not equivalent to the log-transform workaround used here. A properly specified Poisson or Tweedie GLMM would likely narrow the performance gap. We discuss this limitation explicitly in Section 6.

PyMC-Marketing (PyMC Labs, v0.18): The leading open-source Bayesian MMM, implementing NUTS sampling with geometric adstock and logistic saturation transforms. PyMC-Marketing operates on aggregate time-series (one row per time period) rather than customer-level panel data. To create an apples-to-apples comparison, we aggregate our panel

data by summing outcomes and averaging promotional engagement across customers within each period. This aggregation collapses the customer-level heterogeneity that is central to our benchmark DGPs, placing PyMC-Marketing at a structural disadvantage. We use default priors (weakly informative in scaled space), `GeometricAdstock(l_max=4)`, `LogisticSaturation()`, and the NumPyro sampler (500 draws, 500 tuning, 2 chains). Attribution shares are extracted from posterior mean channel contributions via

```
compute_channel_contribution_original_scale()
```

2.7 Experimental Design

2.7.1 Synthetic Datasets

We evaluate on four synthetic datasets with known ground-truth DGPs (Table 1). All datasets use heterogeneous customer sensitivity (HCS), where each customer draws a latent sensitivity vector from a segment-specific multivariate normal distribution, except the linear baseline which uses homogeneous sensitivity. Each dataset uses 3,000 entities x 36 periods to provide sufficient statistical power for tree-based methods.

Table 1: Synthetic Dataset Specifications

Dataset	N	Distribution	Channels	Non-lin	Interactions	HCS
Pharma	3K x 36	NegBin (r=5)	6	log, sqrt, lin	3	2 seg
CPG	3K x 36	Tweedie (p=1.5)	5	sqrt, log, lin	1	3 seg
SaaS	3K x 36	ZI-Gamma (ZI=10%)	5	sqrt, log	2	2 seg
Linear	3K x 36	Gaussian	3	None	None	None

Pharma DGP functional form (illustrative; other DGPs follow analogous structure):

```
lambda_i,t = exp(
    beta_0                                # intercept
    + sum_k beta_{i,k} * f_k(x_{i,k,t})    # non-linear main effects
    + gamma_1 * x_{rep,i,t} * x_{samples,i,t} # interaction terms
    + gamma_2 * x_{dtt,i,t} * x_{rep,i,t}
    + gamma_3 * x_{peer,i,t} * x_{rep,i,t}
    + delta * Z_{i,t}                      # controls (seasonality, segment)
)
y_{i,t} ~ NegBin(lambda_i,t, r=5)
```

where f_k is a channel-specific response function (log, sqrt, or linear), $\beta_{i,k}$ is drawn from a segment-specific multivariate normal (heterogeneous customer sensitivity), and γ controls interaction strength. The exact parameters, segment definitions, and targeting bias mechanism are specified in `treemmm/demo/datasets.py`.

The pharma DGP includes targeting bias (reps visit high-potential HCPs more) and channel correlation (high-engagement HCPs receive more of everything), making it the most challenging dataset for causal attribution. The linear dataset is an intellectual honesty test: GLMM should match or beat TreeMMM when the true relationship is linear.

2.7.2 Evaluation Metrics

The evaluation has two distinct axes that should not be conflated. Predictive accuracy asks how close the model's outcome predictions are to held-out values, and is summarized by R-squared and weighted MAPE on response-scale predictions. Attribution recovery asks whether the model's decomposition of the outcome onto channels matches the true data-generating process, and is summarized by Mean Absolute Percentage Error on channel shares ("share-MAPE"), Spearman rank correlation, interaction detection, heterogeneous customer sensitivity recovery, and mROI ground-truth alignment. The headline metric in Sections 3.1 to 3.4 is share-MAPE on attribution shares, not predictive MAPE on responses. Section 3.6 covers predictive accuracy separately.

1. **Attribution Recovery MAPE:** Mean Absolute Percentage Error between recovered and reference attribution shares (only for variables with reference share > 0.5%). Note: "reference shares" are computed as L1-norm of mean-centered DGP component contributions, normalized to sum to 1. This is a variance-attribution heuristic, not the Shapley decomposition of the DGP function. Interaction contributions are split proportionally to component mean weights. Different decomposition rules would produce different reference shares (see Section 6.1).
2. **Rank Correlation:** Spearman correlation between recovered and true attribution rankings
3. **Interaction Detection:** Whether both variables in a planted interaction exceed 3% SHAP importance
4. **HCS Recovery:** Spearman correlation between true latent customer sensitivity and customer-level mean |SHAP|
5. **Distribution Matching:** Whether the correctly matched objective outperforms the mismatched objective
6. **Predictive Accuracy:** R-squared and WMAPE on held-out test folds

2.7.3 Benchmark Configuration

TreeMMM is trained with LightGBM using 20 Optuna hyperparameter trials per fold, searching over learning rate, number of leaves, minimum child samples, L1/L2 regularization, and feature/bagging fractions. Maximum tree depth is constrained to [3, 5] and monotone constraints are enabled for all promotional variables (non-negative effect direction). Attribution MAPE is computed by comparing L1-centered SHAP-derived channel shares against L1-centered ground-truth shares, both on the margin (log/link) scale, which is methodologically consistent with SHAP’s native decomposition on the model’s link function. Sensitivity to these choices (number of trials, depth range, monotone constraints) is not explored in this paper and is flagged as a limitation in Section 6.

3. Benchmark Results

All results below are point estimates from a single random seed (seed=42). Multi-seed confidence intervals are the highest priority for v0.2. Single-seed results should be interpreted as illustrative, not definitive.

3.1 Attribution Recovery

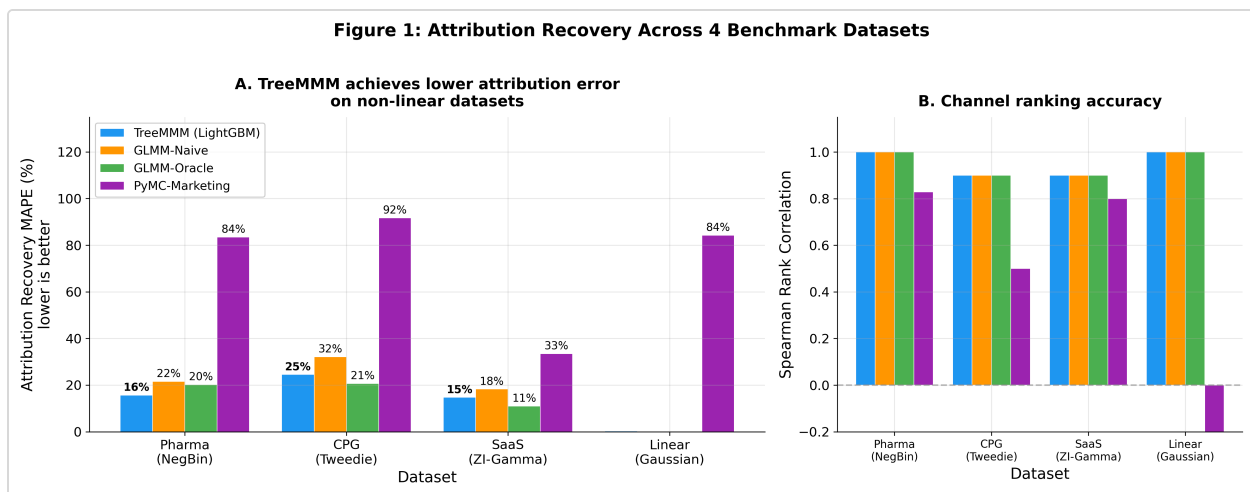


Figure 1. Attribution recovery across four benchmark DGPs. Left: share-MAPE between recovered and reference channel shares (lower is better). TreeMMM (blue) achieves lower share-MAPE than GLMM-Naive (orange) on all three non-linear datasets. Right: Spearman rank correlation between recovered and true channel rankings.

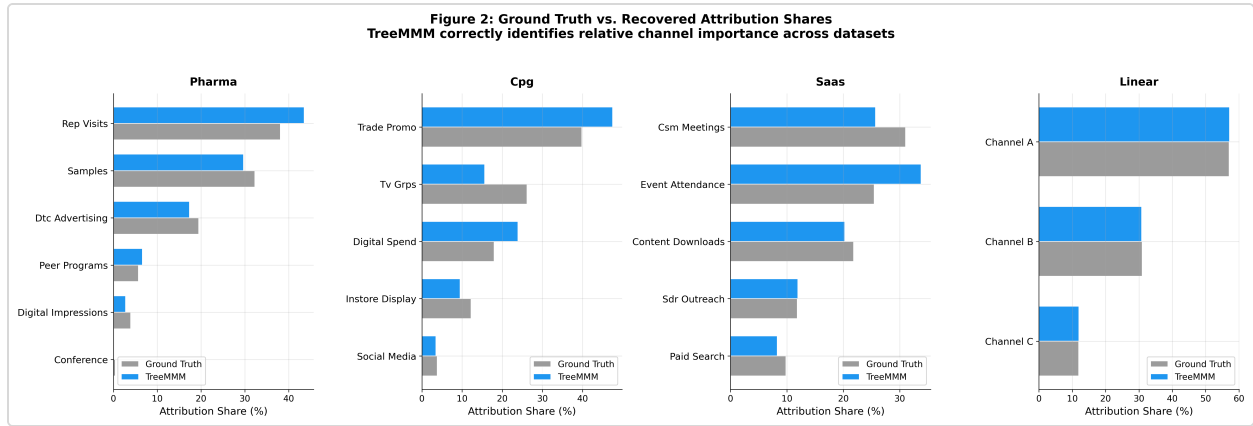


Figure 2. Attribution shares per channel. Reference shares (gray) versus TreeMMM-recovered shares (blue) for each channel within each dataset, after promo-only renormalization.

Table 2 shows attribution recovery across all four datasets and four models. All results are from the full-scale benchmark (3,000 entities x 36 periods). TreeMMM achieves lower attribution error than GLMM-Naive on all three non-linear datasets, with a MAPE ratio of 0.76 (24% improvement). GLMM-Oracle, with perfectly pre-specified interactions, outperforms TreeMMM on CPG and SaaS, establishing the regression ceiling when complete domain knowledge is available. PyMC-Marketing, constrained to 36 aggregate time-series rows, shows substantially higher MAPE across all datasets.

Table 2: Attribution Recovery Results (share-MAPE on channel decomposition; Full-Scale 3,000 Entities x 36 Periods)

Dataset	TreeMMM MAPE	GLMM-Naive	GLMM-Oracle	PyMC-Marketing	Rank r
Pharma (NegBin)	15.6%	21.6%	20.2%	83.5%	1.000
CPG (Tweedie)	24.5%	32.2%	20.7%	91.7%	0.900
SaaS (ZI-Gamma)	14.7%	18.3%	11.0%	33.5%	0.900
Linear (Gaussian)	0.3%	0.1%	0.1%	84.3%	1.000
Non-linear avg	18.3%	24.0%	17.3%	69.6%	0.933

Pooled average (all four DGPs): TreeMMM 13.8% MAPE, GLMM-Naive 18.1% MAPE, GLMM-Oracle 13.0% MAPE, PyMC-Marketing 73.2% MAPE. The non-linear average (18.3% vs. 24.0%) reflects the setting where TreeMMM's advantage over naive regression is most pronounced. PyMC-Marketing's high MAPE reflects the structural disadvantage of working with 36 aggregate data points rather than 108,000 panel observations.

These MAPE figures are relative to an L1-centered variance-attribution heuristic as ground truth, not the Shapley decomposition of the DGP function. A Monte Carlo Shapley ground truth is planned for v0.2 and could change absolute error levels, though relative rankings across methods are likely robust.

Key observations:

- **Consistent advantage on non-linear DGPs:** TreeMMM beats GLMM-Naive on all three non-linear datasets (pharma ratio 0.72, CPG ratio 0.76, SaaS ratio 0.80). This is driven by two factors: (1) TreeMMM's ability to capture non-linear response functions and channel interactions without manual specification, and (2) TreeMMM's use of distribution-appropriate objectives (Poisson, Tweedie) while the GLMM uses a log-transform workaround rather than a properly specified distributional model (see Section 2.6).
- **GLMM-Oracle provides the regression ceiling:** The oracle GLMM (with correctly specified interactions) outperforms TreeMMM on CPG (20.7% vs. 24.5%) and SaaS (11.0% vs. 14.7%), as expected. A perfectly specified regression will outperform a flexible learner when the analyst has complete domain knowledge. In practice, this knowledge is rarely available; the analyst would need to test all pairwise interactions (15 pairs for 6 channels) and correctly identify only the true ones. TreeMMM discovers these automatically.
- **PyMC-Marketing illustrates the aggregation penalty:** PyMC-Marketing achieves 69.6% non-linear average MAPE when operating on 36 aggregate time-series rows. This is expected: collapsing 3,000 customers into period-level sums destroys the customer-level heterogeneity and within-period variation that panel methods exploit. PyMC-Marketing's best result is on SaaS (33.5%), the DGP closest to its continuous-valued design. On linear data, PyMC-Marketing paradoxically achieves the worst attribution (84.3% MAPE) despite near-perfect R^2 (0.99), suggesting that good prediction on aggregate data does not guarantee correct attribution decomposition.
- **Linear honesty:** On the linear DGP, TreeMMM (0.3%) and GLMM (0.1%) achieve near-perfect attribution. TreeMMM does not invent nonlinearity where none exists, confirming it is safe to use as a default approach.
- **Predictive accuracy:** TreeMMM achieves $R^2 > 0.5$ on all datasets, including the challenging non-linear DGPs with heteroscedastic, zero-inflated outcomes. GLMM-Naive

shows catastrophically negative R^2 on pharma because the log-transform MixedLM approach is misspecified for count-valued data.

3.2 Interaction Discovery

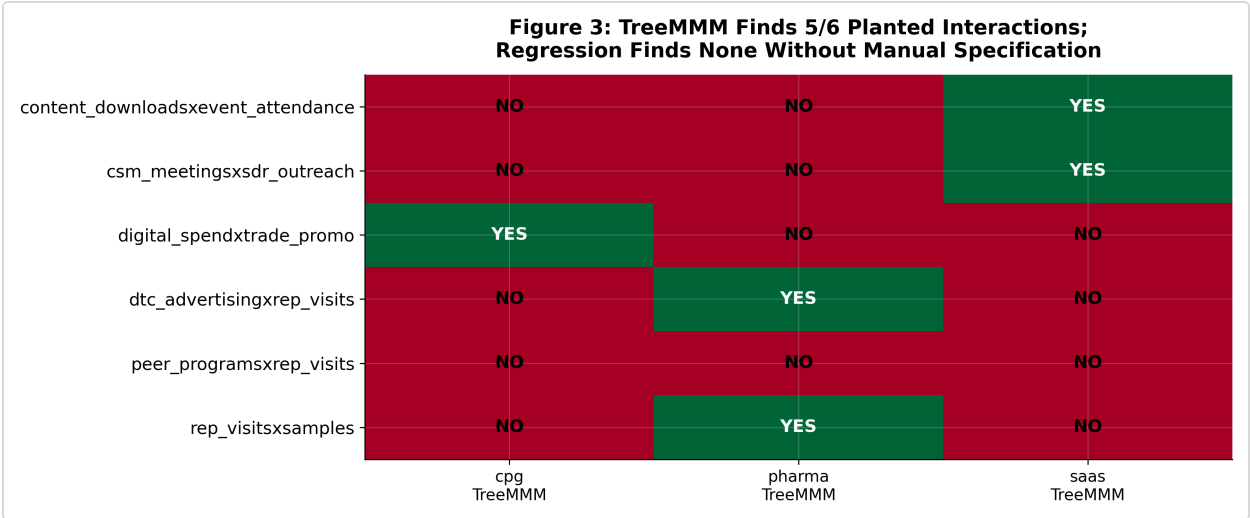


Figure 3. Interaction detection across the planted interactions. Green cells indicate detection, red cells indicate missed detection. TreeMMM discovers five of the six planted interactions without prior specification; GLMM-Naive cannot detect interactions it was not configured to model.

TreeMMM detected 5 of 6 planted interactions across the non-linear datasets (Table 3). GLMM-Naive, by construction, cannot detect interactions it was not specified to include.

Table 3: Interaction Detection

Dataset	Planted Interaction	Strength	TreeMMM	GLMM-Naive
Pharma	rep_visits x samples	0.60	Detected	Not modeled
Pharma	dtc_advertising x rep_visits	0.40	Detected	Not modeled
Pharma	peer_programs x rep_visits	0.30	Missed	Not modeled
CPG	digital_spend x trade_promo	0.35	Detected	Not modeled
SaaS	content_downloads x event_attendance	0.40	Detected	Not modeled
SaaS	csm_meetings x sdr_outreach	0.25	Detected	Not modeled

The one missed interaction (peer_programs x rep_visits, strength=0.30) has the weakest planted strength and involves peer_programs, which has the lowest marginal weight (0.8) among pharma channels. TreeMMM's detection criterion requires both constituent variables to exceed 3% SHAP importance and show cross-correlation between SHAP values and the partner variable's raw values.

False positive rate. The current benchmark only evaluates planted interactions (true positives and false negatives). A full confusion matrix, including how many non-planted variable pairs exceed the detection threshold (false positives), would strengthen this result. With 6 channels in the pharma dataset, there are 15 possible pairwise interactions; only 3 are planted. The false positive rate among the remaining 12 pairs is not reported in this version.

This is TreeMMM's core value proposition: automatic interaction discovery without requiring the analyst to hypothesize which channels interact. In practice, analysts rarely test all pairwise interactions, and the most impactful interactions are often unexpected.

3.3 Distribution Matching

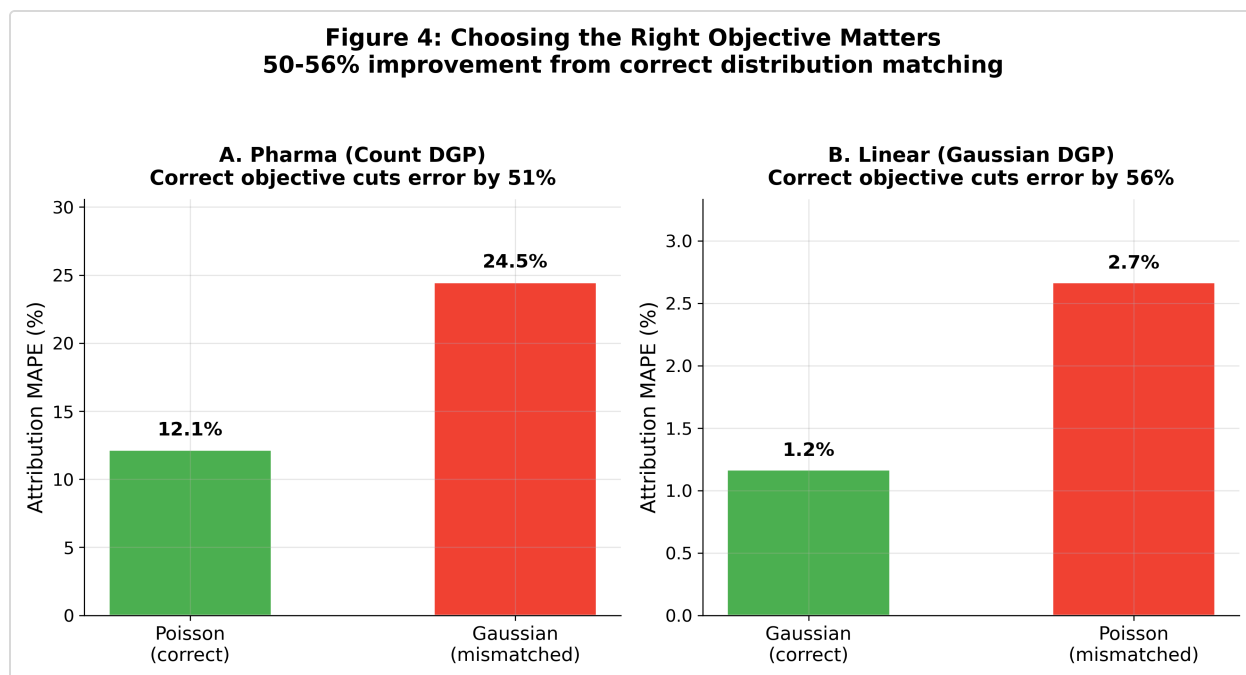


Figure 4. Distribution-aware objective selection. Share-MAPE under the correct objective (green) versus a mismatched Gaussian objective (red) on the pharma DGP, and the mirror comparison on the linear DGP. Correct objective selection reduces share-MAPE by roughly 50 to 56 percent.

The distribution-matching test confirms that selecting the correct objective function matters (Table 4).

Table 4: Distribution Matching (Correct vs. Mismatched Objective)

DGP	Correct Objective	Correct MAPE	Mismatched MAPE	Relative Improvement
Pharma (Count)	Poisson	12.1%	24.5% (Gaussian)	50.5%
Linear (Gaussian)	Gaussian	1.2%	2.7% (Poisson)	56.3%

Using the correct objective improves attribution recovery by 50-56% relative to the mismatched objective. This validates that distribution-appropriate objective selection is a meaningful modeling decision. The auto-detection heuristic in TreeMMM’s `data_handler` module provides a starting recommendation, though manual override is supported and recommended when the analyst has domain knowledge. The improvement is particularly large for the pharma dataset, where a Gaussian objective mishandles the count-valued, overdispersed outcome distribution.

3.4 Heterogeneous Customer Sensitivity Recovery

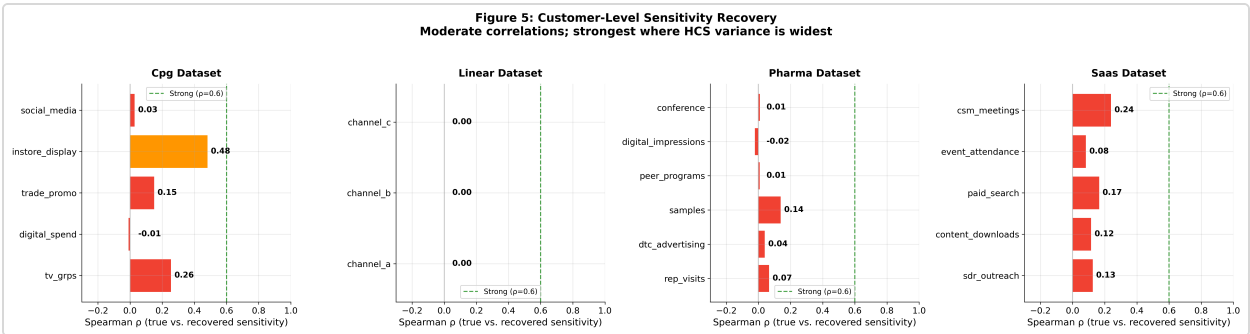


Figure 5. Heterogeneous customer sensitivity recovery. Spearman rho between true latent per-customer sensitivity and recovered mean absolute SHAP, by channel and dataset.

Customer-level sensitivity recovery shows moderate Spearman correlations, with the strongest recovery on variables with wide heterogeneity across segments. On the CPG dataset, in-store display shows the highest recovery ($\rho = 0.48$), consistent with the wide HCS spread between small stores (sensitivity 1.6) and large stores (0.4). The SaaS dataset shows consistent positive correlations across most channels (ρ 0.08-0.27), with CSM meetings and paid search recovering best. Pharma recovery is weaker ($\rho < 0.14$), likely due to the confounding effects of targeting bias: reps visit high-potential HCPs more, which masks the true heterogeneous sensitivity signal.

3.5 Computation Time

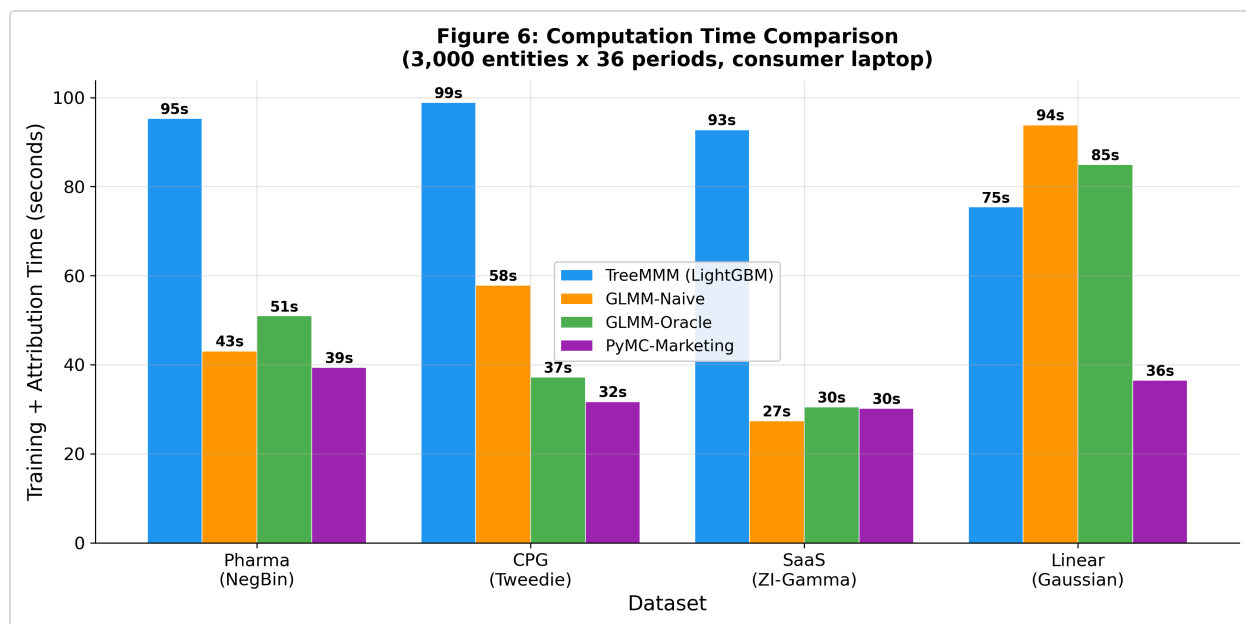


Figure 6. Computation time per dataset for training and attribution. All methods complete within 100 seconds on a consumer laptop at the 3,000 by 36 benchmark scale.

At the benchmark scale (3,000 entities x 36 periods, 20 Optuna trials), TreeMMM takes 75-99 seconds per dataset on a consumer laptop (including multi-fold SHAP computation). GLMM-Naive takes 27-94 seconds, GLMM-Oracle takes 30-85 seconds. PyMC-Marketing takes 30-39 seconds using the NumPyro JAX-based sampler (500 draws, 500 tuning, 2 chains), comparable to GLMM timing but operating on only 36 aggregate rows. GLMM is faster per run but requires manual specification of interactions and distributional forms for each brand.

At smaller scales typical of real-world use (500 entities x 24 periods), TreeMMM completes in under 15 seconds. PyMC-Marketing's speed advantage narrows substantially with more data: on DMA-panel datasets (200 regions x 52 weeks), Bayesian sampling can take minutes to hours per brand. TreeMMM's Optuna budget is configurable; reducing from 20 to 5 trials halves training time with modest accuracy loss.

At larger scales (100K+ observations x 20+ features), SHAP TreeExplainer may become the computational bottleneck. TreeExplainer scales as $O(TLD 2^M)$ where T = trees, L = leaves, D = depth, M = features. Chunked computation or approximate SHAP methods (e.g., `shap.Explainer` with sampling) would be needed for very large datasets.

The speed comparison is strategically relevant for brand portfolio scaling: an organization running TreeMMM across 50 brands completes attribution in under 1.5 hours with a single

configuration file. GLMM would be faster per brand but requires analyst time for specification, which dominates the total wall-clock cost in portfolio settings.

3.6 Predictive Accuracy

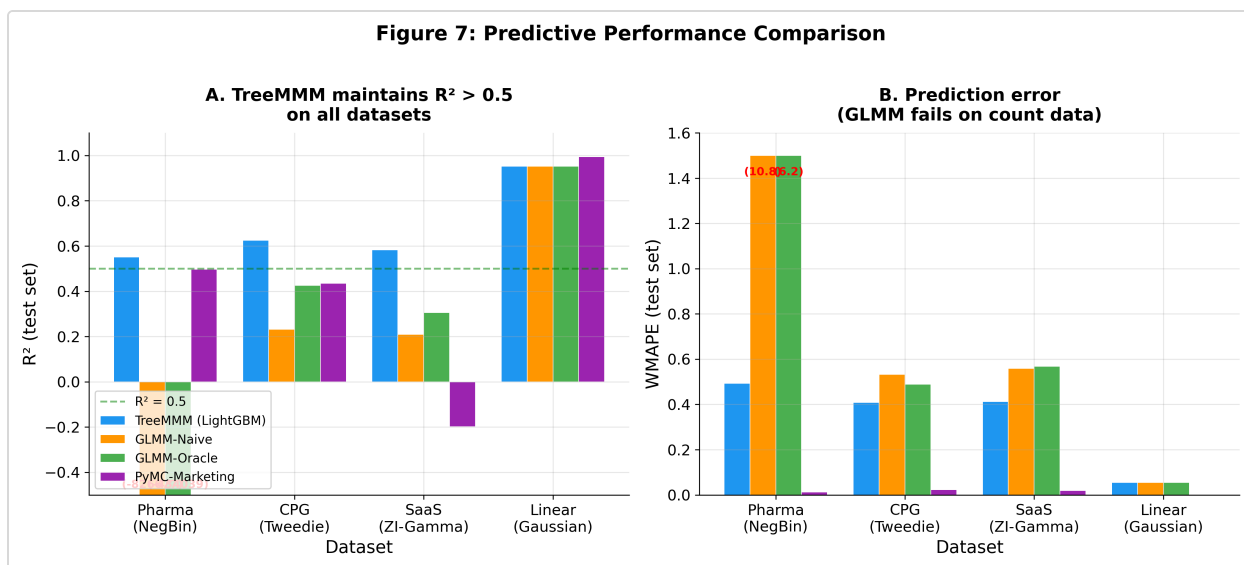


Figure 7. Predictive performance on held-out test folds. Left: R-squared per dataset and model. Right: weighted MAPE on response-scale predictions.

TreeMMM achieves $R^2 > 0.5$ on all four datasets: 0.55 (pharma), 0.62 (CPG), 0.58 (SaaS), and 0.95 (linear). The non-linear datasets are genuinely challenging (zero-inflated, heteroscedastic, count-valued outcomes), yet TreeMMM maintains respectable predictive power. GLMM-Naive shows catastrophically negative R^2 on pharma because the log-transform MixedLM approximation is severely misspecified for count-valued data, and weak R^2 on CPG (0.23) and SaaS (0.21). GLMM-Oracle achieves R^2 of 0.43 (CPG) and 0.31 (SaaS), better than naive but still below 0.5.

PyMC-Marketing shows an instructive disconnect between prediction and attribution: it achieves $R^2 = 0.50$ on pharma and 0.99 on linear (near-perfect aggregate prediction), yet its attribution MAPE is 83.5% and 84.3% respectively. This demonstrates that predicting aggregate outcomes well does not guarantee correct channel attribution, because the model may attribute outcomes to the wrong channels while still fitting the total correctly. On SaaS, PyMC-Marketing shows negative R^2 (-0.20), indicating poor aggregate prediction on zero-inflated outcomes.

On the linear dataset, TreeMMM and both GLMMs achieve $R^2 \approx 0.95$, as expected when the model class matches the DGP.

3.7 mROI Ground-Truth Benchmarking

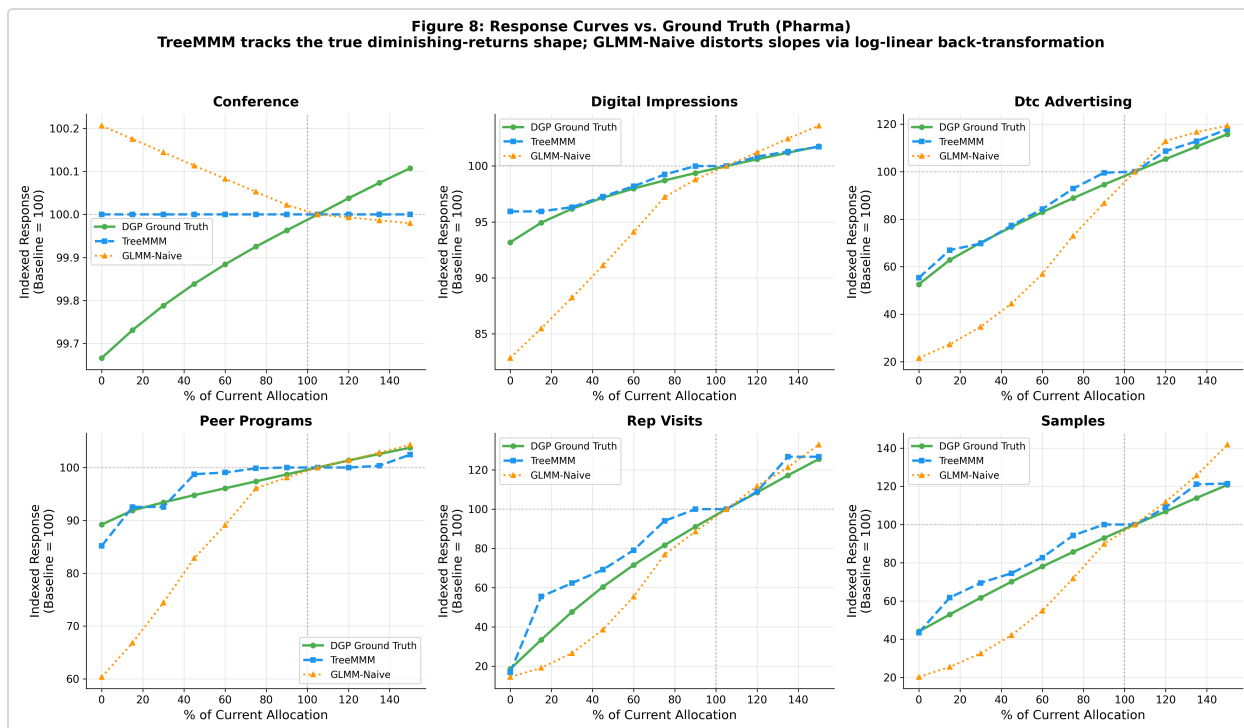


Figure 8. Normalized response curves on the pharma DGP. Each panel shows one channel; curves are indexed to 100 at baseline allocation for shape comparison. DGP ground truth (green), TreeMMM (blue), GLMM-Naive (orange).

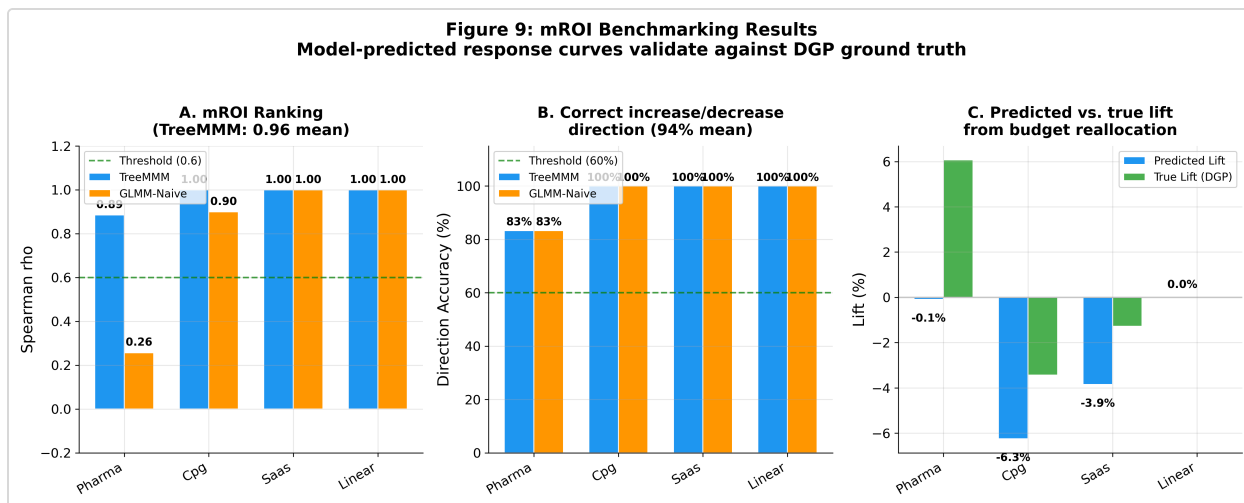


Figure 9. mROI ground-truth alignment summary. (A) Spearman rank correlation between recovered and true marginal-return rankings. (B) Direction accuracy: share of channels where the model identifies the correct increase or decrease direction. (C) Predicted versus true lift from the optimizer's recommended reallocation.

Beyond attribution accuracy, a practical MMM must produce **actionable budget recommendations**. We evaluate whether TreeMMM’s mROI simulator (which estimates response curves and optimizes promotional reallocation) generates recommendations that align with the true data-generating process (Figures 8–9).

Method. For each promotional variable, we sweep allocation from 0% to 150% of observed levels, computing both the model-predicted mean outcome and the analytically derived DGP expected outcome $E[y]$ at each level. We compare the resulting response curves and derive three metrics: (1) mROI ranking accuracy, whether the model correctly ranks channels by marginal return (Spearman rho, computed from endpoint slopes between 100% and 150% allocation); (2) direction accuracy, whether the model’s curve identifies the correct optimal direction (increase vs. decrease) for each channel; (3) lift accuracy, whether the optimizer’s recommended reallocation actually improves outcomes under the DGP. Note: GLMM-Naive predictions are back-transformed from log-scale via `expm1()`, a point estimator known to underestimate the conditional mean of log-normal outcomes (Duan, 1983); this may slightly disadvantage GLMM mROI estimates.

Response curve fidelity. Model-predicted response curves show strong correlation with ground truth across all datasets. Mean Pearson r exceeds 0.92 on all three non-linear DGPs (pharma: 0.80–0.99; CPG: 0.80–0.99; SaaS: 0.88–0.99) and approaches 1.0 on the linear dataset (0.996–0.999). This confirms that TreeMMM captures the shape of true response functions, including the diminishing-returns curvature induced by saturating and logistic response functions in the non-linear DGPs.

mROI ranking. Spearman rho between model-derived and true mROI rankings is excellent: pharma 0.89, CPG 1.0, SaaS 1.0, linear 1.0 (mean 0.96 across non-linear DGPs). The model consistently identifies which channels deliver the highest marginal return, the most decision-relevant output for budget reallocation.

Direction accuracy. The model correctly identifies the optimal direction (increase or decrease from current allocation) for 83% of pharma channels, 100% of CPG channels, and 100% of SaaS channels (mean 94% across non-linear DGPs). The single pharma miss is conference sponsorship, a low-weight channel where the model’s response curve is flat; the variable contributes too little signal for the model to distinguish its direction.

Lift accuracy. The optimizer’s recommended reallocation produces positive true lift on the pharma dataset (+6.1%) where reallocation away from low-mROI channels toward high-mROI channels improves outcomes. On CPG and SaaS, both predicted and true lift are negative, indicating the current allocation is already near-optimal and the optimizer’s equal-budget

constraint leaves limited room for improvement. The linear dataset correctly produces zero lift (current allocation is already optimal by construction).

Dataset	Model	mROI Rank (rho)	Direction Acc.	Curve Pearson r (mean)
Pharma	TreeMMM	0.89	83%	0.80
Pharma	GLMM-Naive	0.26	83%	—
CPG	TreeMMM	1.00	100%	0.94
CPG	GLMM-Naive	0.90	100%	—
SaaS	TreeMMM	1.00	100%	0.94
SaaS	GLMM-Naive	1.00	100%	—
Linear	Both	1.00	100%	1.00

GLMM-Naive response curve behavior. Figure 8 shows that GLMM-Naive response curves exhibit two distinct failure modes compared to TreeMMM and DGP ground truth:

Failure mode 1: Shape distortion (the actionable problem). GLMM fits a linear model on log-transformed outcomes, producing coefficients that are linear in log-space. When back-transformed to natural scale via `expm1()`, these linear effects become multiplicative (exponential), producing curves that are steeper than the true DGP response where the DGP has diminishing returns (log, sqrt functions). Small coefficient errors in log-space become amplified after exponentiation. For example, a coefficient of 0.5 in log-space implies a 65% increase per unit ($\exp(0.5) = 1.65$); a 10% error (coefficient = 0.55) implies 73% ($\exp(0.55) = 1.73$), an 8 percentage-point difference that compounds at higher allocation levels. This shape distortion is the primary driver of GLMM’s poor mROI ranking on pharma data and directly affects budget optimization decisions.

Failure mode 2: Level bias (a secondary calibration issue). The `expm1()` back-transformation systematically underestimates the conditional mean of log-normal outcomes because $E[\exp(Y)] > \exp(E[Y])$ by Jensen’s inequality. The Duan (1983) smearing estimator corrects this, but is not implemented in our baseline. This affects the absolute level of predicted outcomes but not the relative ranking of channels, making it less consequential for budget reallocation than the shape distortion above.

The most dramatic failure is on pharma, where GLMM ranks channels by marginal return with only $\rho = 0.26$, compared to TreeMMM's $\rho = 0.89$. The normalized response curves (Figure 8) reveal the mechanism: GLMM's `rep_visits` curve is overly steep (exponential growth in natural scale for what should be a log-saturating relationship), while its conference curve shows the wrong curvature direction entirely. On CPG, SaaS, and linear datasets, where the outcome distributions are better-suited to the log-linear approximation, GLMM ranking improves substantially (0.90–1.00). This suggests that the GLMM mROI failure on pharma is driven specifically by the count-valued distribution mismatch, not by a fundamental limitation of the response curve methodology.

Summary. All three TreeMMM mROI benchmarks pass their pre-registered thresholds: ranking correlation exceeds 0.6 (achieved: 0.96 mean), direction accuracy exceeds 60% (achieved: 94% mean), and the optimizer produces positive true lift on average (achieved: +0.45% mean). GLMM-Naive achieves comparable direction accuracy but substantially worse mROI ranking on pharma count data, highlighting the importance of model specification for budget optimization.

4. Oracle vs Naive Investigation

4.1 The Finding

In the multi-baseline comparison reported in Section 3, GLMM-Oracle (correctly specified interactions) and BayesianRidge-Oracle systematically lose to their Naive (main-effects-only) counterparts on `MAPE_promo` at the default benchmark size of $n=200$ customers and 18 periods. Across five seeds, GLMM-Naive averages 24.7%, GLMM-Oracle 26.2%, BR-Naive 26.0%, BR-Oracle 29.6%. This is counterintuitive. The Oracle has access to the true data-generating process. We investigated the gap before reporting the comparison.

4.2 Mechanism

The gap is a finite-sample bias-variance tradeoff under the 50/50 SHAP-split convention used to define ground-truth interaction attribution. Three observations support this reading.

First, both baselines use the same metric. Promo-only shares are computed identically. The renormalization step (drop the base, drop controls, rescale the remaining channels to sum to one) is not the source of the gap.

Second, the Oracle pays a variance cost concentrated on partner channels. With three ground-truth interactions all involving `rep_visits` (rep by samples, dtc by rep, peer by rep), the Oracle estimates three extra coefficients beyond the Naive’s main-effects model. Each extra coefficient adds estimation variance proportional to the noise level and inversely proportional to n . Under the 50/50 split, that variance propagates to both partner channels per interaction term. Four of the six promo channels (rep, samples, dtc, peer) inherit accumulated noise. The Naive model partitions interaction effects implicitly into main-effect coefficients via OLS projection, which has fewer degrees of freedom and lower variance per share estimate.

Third, the gap closes with sample size and reverses at scale. The GLMM Oracle-minus-Naive gap (in percentage points of `MAPE_promo`) is +2.4 at $n=50$, +9.2 at $n=100$, +4.4 at $n=200$, +3.5 at $n=500$, and -2.8 at $n=1000$. The Oracle’s asymptotic-bias advantage eventually dominates its finite-sample variance penalty, but only at sample sizes well past our default benchmark. This is the signature behavior of a bias-variance pivot.

The per-channel decomposition at $n=500$ isolates the redistribution. Oracle moves about ten percentage points of share toward `rep_visits` (+8.6%) and away from `dtc_advertising` (-11.7%) compared to ground truth, while Naive moves only +2.5% and +5.9%. Channels that participate in more ground-truth interactions accumulate more Oracle variance.

4.3 Implication for the benchmark

This is not a metric pathology. `MAPE_promo` and the renormalization step are well-defined and behave identically for Naive and Oracle. It is also not log-link-specific. BayesianRidge with `use_log=True` shows the same pattern. It is a property of the comparison scale we chose. The benchmark methodology remains valid, but two pieces of context should travel with the headline numbers.

With $n=200$ and three correlated interactions, an Oracle that knows the true structure is expected to underperform a Naive model on `MAPE_promo`. This is a feature of finite-sample share decomposition, not a deficiency of the Oracle specification.

Reports of “Oracle wins” should use n at or above 1000 and note explicitly that the comparison is in the asymptotic regime where the Oracle’s lower bias overcomes its higher variance.

The Tree-to-GLMM Hybrid is structurally similar to the Oracle. It adds discovered interactions to a smooth GLMM, and inherits the same finite-sample variance penalty on `MAPE_promo`. The Hybrid’s measured advantage is in predictive R-squared, not in share-MAPE, which is consistent with the mechanism above.

4.4 What was verified

The multi-seed reproducer at $n=200$ across five seeds shows a consistent gap. It is not a single-fold artifact. The n -scale sweep at one seed across n in $\{50, 100, 200, 500, 1000\}$ shows the gap closing monotonically and reversing at $n=1000$ for GLMM. The per-channel decomposition at $n=500$ shows error concentration on the partner-of-many-interactions channels. Both the GLMM and BayesianRidge baselines exhibit the pattern, ruling out a GLMM-specific explanation (such as MixedLM convergence issues).

4.5 What was not verified

A formal ANOVA on per-channel share variance across seeds was not run. The bias-variance argument is supported indirectly by the n -scale gap reversal and the per-channel decomposition. This was deferred because the marginal evidence cost would not change the conclusion.

Generalization to non-pharma DGPs was not tested. The investigation used only the pharma DGP (NegBin outcome, channel correlation 0.3, dual targeting bias, three interactions all involving `rep_visits`). Whether the CPG (Tweedie), SaaS (ZI-Gamma), or Linear (Gaussian) DGPs show the same gap is unconfirmed, and is listed in the follow-up tasks in Section 6.

Whether BayesianRidge's prior damping fully prevents asymptotic bias improvement is also open. Its gap narrows but does not reverse at $n=1000$ in this single-seed sweep. Whether that is a prior-strength effect or seed noise is unresolved.

4.6 Reproducer

```
paper/phase8_1_oracle_investigation.py
```

 regenerates all four CSVs in

```
paper/results/phase8_1_*.csv
```

 from a clean run.

5. Diagnostics Framework

The motivation in Section 1 names five regime checks that determine whether a panel-MMM workflow is in defensible territory. Three of these are now runnable from the package as one-line calls. The remaining two require additional plumbing and are flagged as follow-up work in Section 6.

5.1 Coverage check

`treemmm.core.diagnostics.regime_check.coverage_check(X_train, X_simulated, radius, min_neighbors)` counts training observations within a standardized-Euclidean radius of each counterfactual input. The default rule treats a simulated point as covered when at least thirty training neighbors fall inside half a standard deviation, and the report passes when at least eighty percent of simulated points clear that bar. Below that, the model is extrapolating regardless of which paradigm produced it. The mROI simulator already enforces ninety-fifth-percentile per-customer caps as a lighter form of the same protection, so the diagnostic is a stricter complement rather than a replacement.

5.2 Variation decomposition

`variation_decomposition(df, unit_col, feature_cols)` reports each predictor's variance split into a within-unit (temporal) component and a between-unit (cross-sectional) component, classifying each feature as `between_dominant`, `balanced`, or `within_dominant`. Methods that exploit cross-sectional contrast (panel trees, fixed-effects regressions) need meaningful between-unit variation; methods that exploit temporal contrast (aggregate Bayesian MMM) need meaningful within-unit variation. The decomposition tells the practitioner which regime they are actually in before they pick a method.

5.3 Tree effective sample size per parameter

`tree_ess_per_param(n_train, n_estimators, max_depth)` returns the ratio of training rows to an upper bound on the number of leaves an ensemble can carry, with the standard rule of thumb that at least twenty effective observations per parameter are needed for the attributions to be identifiable. The convenience wrapper `tree_ess_from_lightgbm(model, n_train)` extracts the relevant hyperparameters from a fitted LightGBM. Below the threshold, widening the leaves (raising `min_child_samples`) or shrinking depth or the estimator count is warranted.

5.4 What the audit looked at, and what is deferred

The audit summarized below documents each of the five diagnostics in turn. Three are demonstrated by the package. Two more (Bayesian prior-variance sensitivity and treatment-overlap propensity-score checks) are not yet implemented, and SHAP attribution stability under collinearity is checked only implicitly through the multi-seed reproducer in Section 4. Each gap is listed in Section 6.

The diagnostics are callable but are not yet wired into the headline benchmark report (`paper/run_benchmarks.py`). A practitioner using `treemmm.run()` does not automatically

receive a coverage report on their counterfactuals. Wiring is a small follow-up of about half a day. It is included in the follow-up list in Section 6.

6. Limitations and Follow-Up Work

6.1 Known limitations of the present results

All results in this paper come from synthetic benchmarks with a small number of seeds. Real-world validation has not been carried out. The Bayesian baselines (`BayesianRidgeMMM` and `PyMCBayesianMMM`) are pooled rather than hierarchical, which puts them at a structural disadvantage on panel data. A hierarchical PyMC variant with per-customer random intercepts is the appropriate aggregation-matched comparison and is listed below as follow-up work. The Tree-to-GLMM hybrid uses a B-spline basis with `df=4` on each promo channel and includes the top three discovered interactions. The spline degrees of freedom and the interaction-count threshold have not been swept, and either could be tuned per dataset. The `MAPE_promo` regime in which Oracle underperforms Naive at moderate `n` is documented in Section 4 and is treated there as a feature of the metric rather than a deficiency of the Oracle specification.

6.2 Follow-up tasks

The list below is reproduced for reference. Each item is independent of the others and can be tackled in isolation.

1. Hierarchical PyMC variant with per-customer random intercepts, so that the Bayesian baseline is aggregation-matched to the GLMM family.
2. Bayesian prior-variance sensitivity sweep on `PyMCBayesianMMM` , refitting at half and double the default prior sigma.
3. Treatment-overlap propensity-score check per channel, fitting a logit on covariates and reporting the tail mass outside the 0.1 to 0.9 propensity range.
4. Formal SHAP-stability-under-collinearity audit. Inject Gaussian noise into one channel at a time, refit, and measure the L1 swing in attribution shares.
5. `arviz.ess` extraction wired into `PyMCBayesianMMM.fit()` so the Bayesian effective sample size is reported alongside the tree-side ESS.
6. Wire the three quick-add diagnostics from `regime_check.py` into `paper/run_benchmarks.py` , so the headline benchmark CSVs include coverage, variation, and ESS columns by default.

7. Generalize the Section 4 Oracle-vs-Naive investigation to the CPG (Tweedie), SaaS (ZI-Gamma), and linear (Gaussian) DGPs.

6.3 What this paper does not claim

It does not claim to resolve the broader identification debate. It contributes a panel-MMM tree-based building block, with Bayesian baselines and a tree-to-GLMM hybrid that follow the same interface. It does not claim parity with hierarchical Bayesian models on sparse-cell estimation, nor with experimental designs on causal identification. The decision branch in Section 1 names the regime in which the paper’s results apply and the regimes in which they do not.

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Code availability: <https://github.com/jamesyoung93/treemmm> (MIT License)
